

ARFAT NAIK

Data Science & Machine Learning Aspirant

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SUMMARY

Aspiring Data Scientist with hands-on experience in end-to-end machine learning projects, model deployment, and applied NLP. Skilled in Python, scikit-learn, Pandas, and Flask, with demonstrated results on real-world datasets. Seeking a Data Science / Machine Learning internship.

TECHNICAL SKILLS

Programming & Data: Python (NumPy, Pandas) , SQL

ML: Supervised Learning (Regression, Classification), Feature Engineering, Model Training & Model Evaluation, Cross-Validation, Hyperparameter Tuning, Performance Metrics (ROC-AUC, Precision, Recall, F1-Score)

Data Science: Exploratory Data Analysis (EDA), Data Visualization (Matplotlib, Seaborn), Statistical Analysis, Data Cleaning, Outlier Detection, Missing Value Imputation

Tools & Frameworks: Git, GitHub, Flask, Joblib, VS Code, Jupyter Notebook

NLP / Recommendation Systems: TF-IDF Vectorization, Cosine Similarity, Content-Based Recommendation Systems

PROJECTS

Credit Risk Prediction System

- Developed a credit risk prediction ML system using **Logistic Regression** and **cost-sensitive thresholding** to optimize loan approval decisions.
- Engineered features, prevented data leakage, **imputed missing values**, and **managed class imbalance** for robust model evaluation.
- Deployed model via **Flask** for real-time default probability estimation and decision output.

Repository: https://github.com/Arfatnaik0/credit_risk_prediction_system

Movie Recommendation System.

- Developed a content-based recommender for **5,000+ TMDb** movies using **TF-IDF** and **cosine similarity**.
- Engineered an NLP feature pipeline that combines genres, keywords, cast, director, and overview.
- Built Flask web app with **real-time recommendation API** and input validation.
- Optimized similarity computation using sparse matrices, reducing runtime by **60%**

Repository: https://github.com/Arfatnaik0/movie_recommendation

Bangalore House Price Prediction

- Trained regression models on 13,000+ listings, achieving MAE of **22-23 Lakhs** INR.
- Performed data preprocessing, including missing value imputation, **IQR-based outlier removal**, and **categorical encoding**.
- Evaluated **Linear Regression vs Random Forest** using MAE, RMSE, and R2 Score.
- Deployed **Flask-based prediction system** with input validation and error handling.

Repository: https://github.com/Arfatnaik0/bangalore_house_price_prediction

EDUCATION

Bachelor's Degree in Electronics and Computer Engineering

Rizvi College of Engineering (2023-2027)

Semester 6 | CGPA: 7.27

Certifications

Programming, Data Structures and Algorithms using Python – NPTEL

Certificate- <https://drive.google.com/file/d/1-6FNNG1wNs6x1pBJ9LSpyiDSFKkMzJts/view?usp=sharing>